# PROJECT DESIGN PHASE

Project Proposal • Problem–Solution Fit • Solution Architecture • Dashboard Design

| | | | |
|---|---|---|---|
| **Student** | Harshvardhan Gupta | **Course** | Bachelor of Computer Applications (BCA) |
| **Branch** | Computer Applications | **Year / Semester** | Second Year / 3rd Semester |
| **College** | MEDI-CAPS University | **Enrollment ID** | en25ca3020071@medicaps.ac.in |
| **Project Type** | Individual Project | **Maximum Marks** | As specified in each activity template |
| **Project Title** | Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report | **Purpose** | To learn Tableau and data analytics |

## 1. Project Proposal (Proposed Solution)

### Objective

Develop practical knowledge of Tableau and data analytics by exploring global fuel prices, taxation, subsidy levels, regional differences, income groups, and historical patterns through comparative visual analysis and interactive dashboards.

### Problem Statement

| Resource | Project allocation / notes |
|---|---|
| Data | global_fuel_prices_Dataset.csv; approximately 27,468 records, 84 countries and 7 regions, with coverage described as January 2020–April 2026. |
| Software | Tableau Public 2026.3 and a modern web browser. |

| | |
|---|---|
| Hardware | A computer capable of running Tableau Public and a browser; exact device specifications were not recorded. |
| Storage / hosting | Local CSV and workbook files; published workbook hosted on Tableau Public. |
| Personnel | Individual project completed by Disha Patidar. |

Comparing fuel prices across countries is difficult when fuel-price, tax, subsidy, region, income-level, and date information is viewed only as a large raw dataset. Without organized visual analysis, users may struggle to identify patterns and compare groups consistently.

### Scope

The project uses the supplied CSV dataset and Tableau Public 2026.3. It covers country and regional comparisons, taxation and subsidy patterns, income-level comparisons, price rankings, petrol/diesel comparisons, and historical fuel-price trends alongside Brent crude. The project does not provide real-time prices, forecasts, or proof of causal relationships.

### Proposed Solution

Create and publish a Tableau Public workbook with three scenario-based dashboards: (1) Fuel Taxation & Subsidy Policy; (2) Global Fuel Prices & Supply Chain Planning; and (3) Fuel Prices, Taxation & Economic Development. A historical trend worksheet supports time-based exploration.

### Key Features

| Feature | Purpose |
|---|---|
| Country and regional comparisons | Compare average fuel prices and tax measures across countries and regions. |
| Rankings | Identify top and bottom countries by average petrol price or tax percentage where shown. |
| Taxation and subsidy views | Explore how tax percentages and subsidy categories vary alongside fuel prices. |
| Income-level comparisons | Compare average petrol prices and average tax percentages across income groups. |
| Historical trend view | Review fuel-price trends alongside Brent crude over the available time period. |

### Resource Requirements

## 2. Problem–Solution Fit

This section maps the project's intended users and their analytical needs to the capabilities of the Tableau workbook. The personas are project-defined and inferred; they are not presented as findings from interviews or formal user testing.

| Canvas element | Project-specific analysis |
|---|---|
| Customer segments | Energy policy consultants; global supply-chain managers; energy economics researchers, students, and other users comparing fuel-price and taxation patterns. |
| Jobs to be done | Compare taxes and petrol prices by country/region; identify higher- and lower-price countries; explore price differences by subsidy and income level; inspect historical price movement alongside Brent crude. |
| Triggers | Preparing a policy comparison, assessing fuel-cost differences across markets, explaining regional variation, or exploring how prices and taxation differ over time. |
| Before / after | Before: the raw dataset may feel difficult to scan. Intended after-state: charts, rankings, and filters make selected comparisons easier to explore. This is an expected benefit, not a measured outcome. |
| Existing alternatives | Manual CSV/spreadsheet review, individual country statistics, published reports, and other data visualizations. |

| | |
|---|---|
| Constraints | Different data-literacy levels, limited time, screen-size constraints, unfamiliarity with Tableau, and the need to understand data coverage and definitions. |
| Expected behaviour | Open the published workbook, select a relevant dashboard, read chart titles and legends, compare countries or groups, use available filters, and interpret patterns carefully. |
| Channels | Tableau Public in a web browser; workbook link shared for viewing; observations may be discussed in a class presentation or report. |
| Root cause | Fuel-price, tax, subsidy, region, income-level, and date fields are harder to compare consistently when presented as a large table. Different scales and category groupings can obscure patterns. |
| Solution | A Tableau Public workbook containing three dashboards and a supporting historical trend view, using country rankings, regional comparisons, taxation/price views, subsidy and income-level comparisons. |
| **Fit summary** | The workbook aligns with the identified analytical tasks by consolidating relevant measures into comparative charts and trend views. Fit is reasoned from the project requirements and implemented features; it has not been validated through interviews, surveys, or formal usability testing. |

# 3. Solution Architecture

The architecture describes the actual CSV-to-Tableau workflow. No separate database, external ETL pipeline, live price API, or automated refresh is assumed.

| Component | Role in the solution | Technology |
|---|---|---|
| Data source | Historical observations for fuel prices, tax percentage, region, subsidy level, income level, country and date. | Supplied CSV dataset |
| Data connection | Connects the source file to the workbook. | Tableau Public data connection |
| Data preparation | Field types/roles, filters, groupings and built-in aggregations as configured. | Tableau Public 2026.3 |
| Analysis logic | Measures and comparisons used by the worksheets; calculated fields only where configured. | Tableau worksheets / calculations |
| Visualization layer | Charts for tax, petrol/diesel, subsidy, income-level, country ranking and historical trends. | Tableau worksheets |
| Dashboard layer | Organizes the analysis into three scenario-based dashboards. | Tableau dashboards |
| Publishing and access | Makes the workbook available for browser-based exploration. | Tableau Public |
| End user | Views charts, legends and available filters to explore historical patterns. | Web browser |

### Data Flow

| 1. Source | 2. Connect & prepare | 3. Analyse | 4. Present | 5. Publish |
|---|---|---|---|---|
| CSV dataset | Tableau field settings, filters and aggregation | Worksheets and chart comparisons | Three dashboards and trend view | Tableau Public viewer |

### Architecture and scope limitations

The workbook analyses the supplied historical dataset, with coverage documented as approximately January 2020 to April 2026. It does not automatically fetch current pump prices or update itself through an external API. Charts support descriptive comparison; associations between taxes, subsidies, income, crude oil and fuel prices do not establish causation. The supporting worksheet “Global Petrol Price Trend” is not counted as one of the three dashboards.

## 4. Dashboard Design

Dashboard design emphasizes readable layout, appropriate visual encodings, consistent labels and units, and filters that help viewers focus on a specific question. The following design descriptions are aligned to the three dashboards in Disha’s workbook.

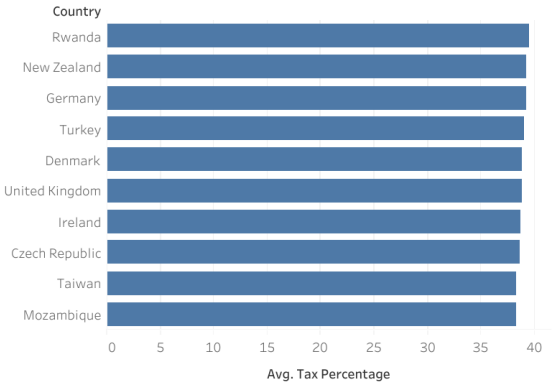
### Dashboard 1 — Fuel Taxation & Subsidy Policy

| Visual / component | Design purpose | Interpretation guidance |
|---|---|---|
| Top 10 Countries by Average Fuel Tax Percentage | Identify nations with the highest fuel tax burdens. | Spot the top tax tier and note the highest percentage baseline. |
| Impact of Taxation on Petrol Prices | Show how tax rates correlate with retail pump prices. | Look for a positive trend and identify any pricing outliers. |
| Average Petrol Price by Region (USD/Liter) | Compare fuel costs across global geographical zones | Pinpoint the most and least expensive continental regions. |
| Average Fuel Prices by Subsidy Level | Assess how different subsidy levels impact market fuel prices. | Compare "Very High" vs "Low" subsidy prices across fuel types.. |

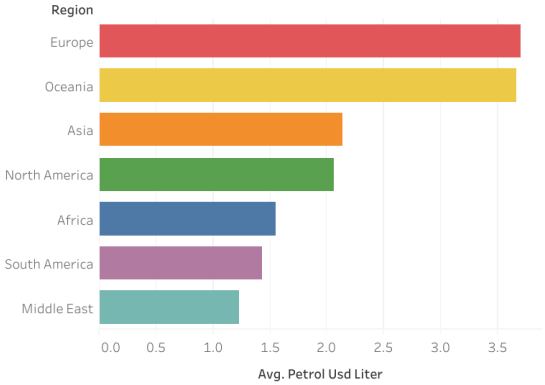
**Scenario 1 - Fuel Taxation & Subsidy Policy**
**An analysis of fuel prices, taxation, and subsidies to support effective energy policy decisions.**



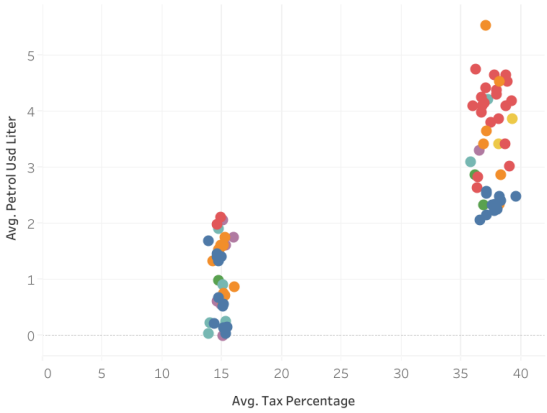

*Top 10 Countries by Average Fuel Tax Percentage*



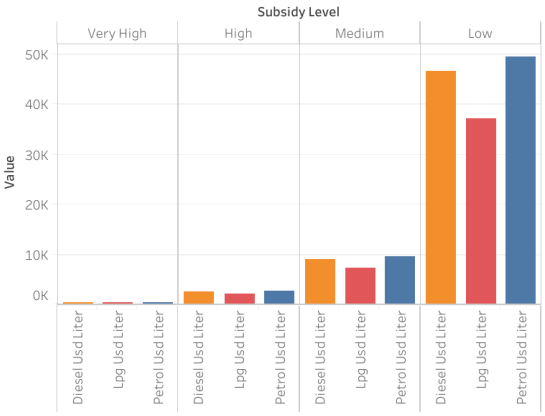

*Average Petrol Price by Region (USD/Liter)*




*Impact of Taxation on Petrol Prices*




*Average Fuel Prices by Subsidy Level*



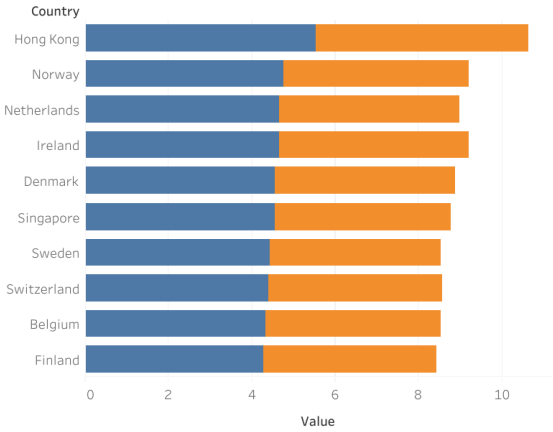
## Dashboard 2 — Global Fuel Prices & Supply Chain Planning

| Visual / component | Design purpose | Interpretation guidance |
|---|---|---|
| Average Petrol Price by Region (USD/Liter) | Establish regional cost benchmarks for transit planning. | Target low-cost regions for sourcing and highcost zones for risk. |
| Top 10 Countries by Average Fuel Price Compare | Highlight the most expensive nations for fuel sourcing. | Compare cost splits between Diesel, LPG, and Petrol in high-cost areas. |
| Bottom 10 Countries by Average Fuel Price Compare | Identify the cheapest global corridors for logistics savings | Find which specific fuel drive the lowest prices in these areas. |
| Brent Crude Oil vs. Regional Petrol Price Trends | Track how retail prices respond to crude oil fluctuations. | Monitor market price sync and check for lag time during shifts. |

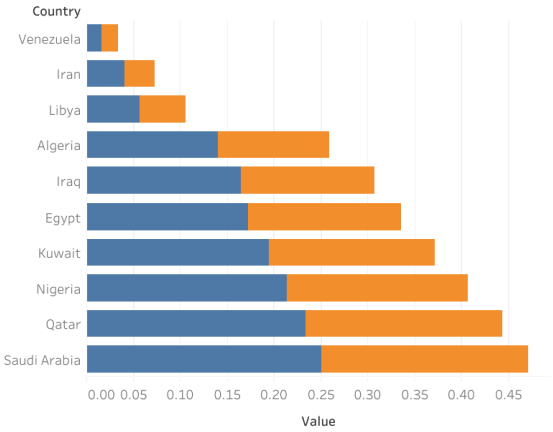
**Scenario 2 - Global Fuel Prices & Supply Chain Planning**
**Analysing global fuel prices to support cost-effective transportation and supply chain decisions.**



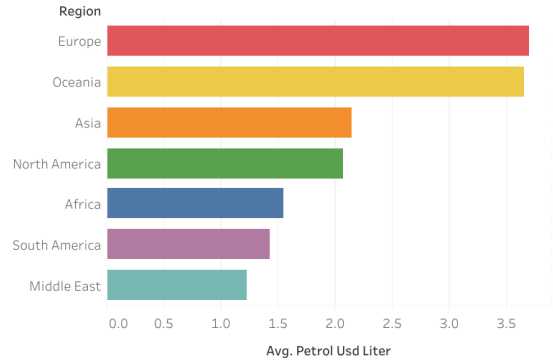

*Top 10 Countries by Average Fuel Price Compare*



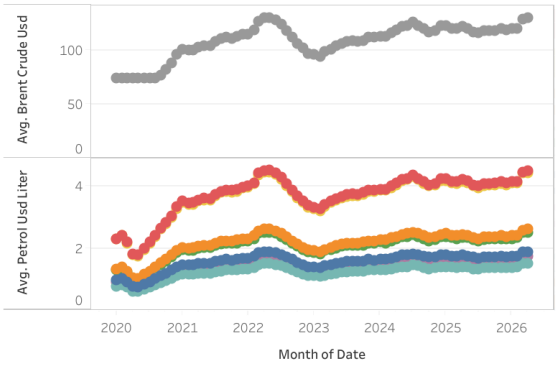

*Bottom 10 Countries by Average Fuel Price Compare*




*Average Petrol Price by Region (USD/Liter)*




*Brent Crude Oil vs. Regional Petrol Price Trends*

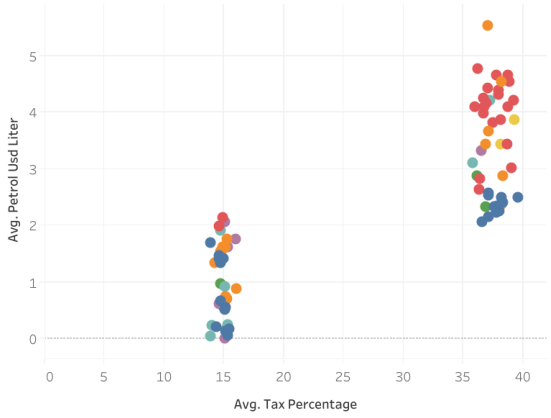


## Dashboard 3 — Fuel Prices, Taxation & Economic Development

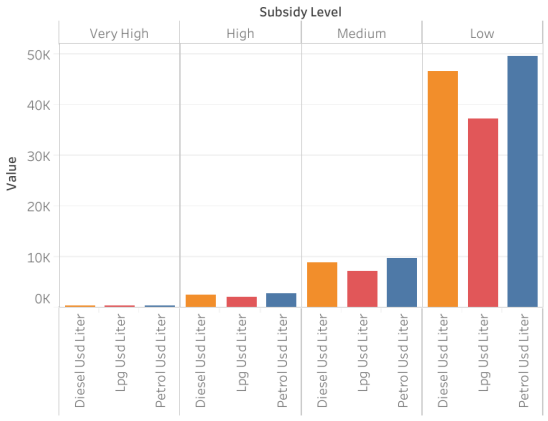
| Visual / component | Design purpose | Interpretation guidance |
|---|---|---|
| Impact of Taxation on Petrol Prices | Check if tax-to-price patterns change across wealth tiers. | Observe if tax changes hit specific income groups harder. |
| Average Fuel Price by Income Level | Compare fuel prices across Low, Middle, and High-income nations. | See if higher-income economies generally face higher fuel costs. |
| Average Fuel Tax Percentage by Income Level | Analyse tax policy reliance based on economic development. | Determine if high-income nations levy larger tax percentages. |
| Average Fuel Prices by Subsidy Level | Evaluate how subsidies cap prices across economic tiers. | Track which fuel types benefit most from artificial caps.. |

**Scenario 3 - Fuel Prices, Taxation & Economic Development**
**Examining fuel prices, taxation, and subsidies across economies at different income levels.**

*Impact of Taxation on Petrol Prices*



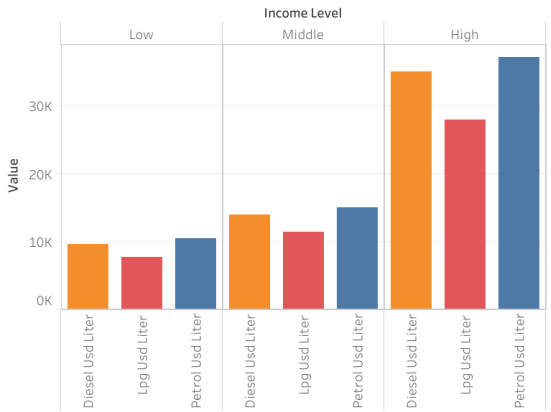

*Average Fuel Prices by Subsidy Level*



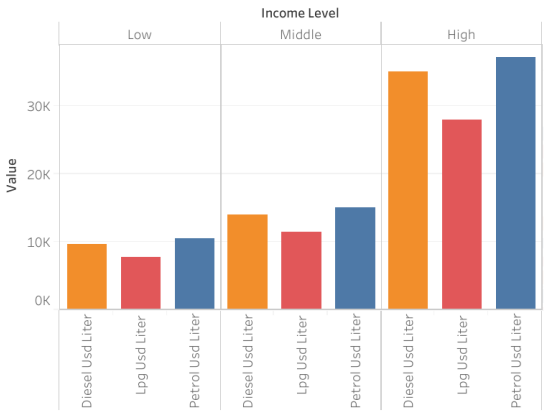




*Average Fuel Price by Income Level.*




*Average Fuel Tax Percentage by Income Level*







# Dashboard Outcomes and Publishing

## Major outcomes

| Outcome | What the dashboard helps the viewer explore |
|---|---|

| | |
|---|---|
| Policy comparison | Country and regional differences in fuel-tax percentages, petrol prices, and subsidy categories. |
| Supply-chain context | Higher- and lower-price country rankings, petrol/diesel comparisons, and historical trend context. |
| Economic-development exploration | Average petrol prices and tax percentages across income groups, plus subsidy-level comparisons. |
| Interactive exploration | Viewers can inspect chart details and use legends or available filters included in the published workbook. |
| Learning outcome | The project provides practice in data connection, aggregation, visualization, dashboard composition, and public publishing in Tableau. |

## Publishing and project links

**Tableau Story (main project entry point):**
**https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Story1**

**Project website:** https://fuel-prices-taxation-dashboard.onrender.com/

**GitHub repository:** https://github.com/Harshvardhan-30/global-fuel-prices-taxation-analysis/tree/main

| | |
|---|---|
| **Design and evidence note** | The dashboard descriptions document intended analytical use and known workbook components. They are not a substitute for a fresh visual audit of every published chart. Use the Tableau Story URL as the general project link; a URL ending in an individual worksheet name should only be used when linking to that specific chart. |